Lightweight CNN based on Non-rigid SFM for 3D Reconstruction of Medical Images

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***Abstract*—We design a lightweight CNN, a practical and inexpensive scheme to reconstruct the geometry of the medical target from a 2D medical sequence. We make full use of natural images with rich texture features to compensate for the low-quality problem of fewer texture conditions in medical images. Besides, we design a Non-rigid Structure from Motion(NRSfM) scheme to estimate the camera structure and pose of consecutive frames with the view synthesis as the supervisory signal. Specifically, for achieving 3D reconstruction with a better time performance, we compress the CNN and medical image 3D prediction can be achieved in the lightweight CNN with the model size of 18M. The experiment results from LITFL of the open clinical library, as well as the Synapse multi-organ segmentation dataset, show that the proposed method can reconstruct the accurate geometry with better time performance.**

Keywords—Non-rigid Structure from Motion, medical image, 3D reconstruction, lightweight CNN

I. INTRODUCTION

Over the past decade, 3D reconstruction has made great achievements in the rapidly developing subfield of computer vision [1-4]. SFM based reconstruction recoveries photogrammetric parameters based on the extracted features [5] without specifying any prior conditions, so it has attracted wide attention [6,7]. But medical images usually have low-quality texture or severe speckle noise [8-12], which makes it difficult to obtain robust features in reconstruction. Unlike medical images, natural images contain higher quality and richer textures, which can provide rich texture to infer stronger perceptual features, therefore, more effective reconstruction results have been obtained.

Recent development of deep learning, knowledge transfer has been proved to be effective in medical image analysis [13-16].Encouraged by the existing works [17,18,19], in our previous work [20], we proposed the SFM-based framework and studied a transfer learning strategy for medical images reconstruction. On this basis, we introduce a teacher-student framework to deal with the time performance problem and provide a lightweight CNN based on the teacher-student strategy. Fig.1 are the results with our models with different scales.

Compared with the previous 3D construction, FlowNet [21,22] takes the original three consecutive frames as input and the GeoNet [23] which only focuses on the rigid

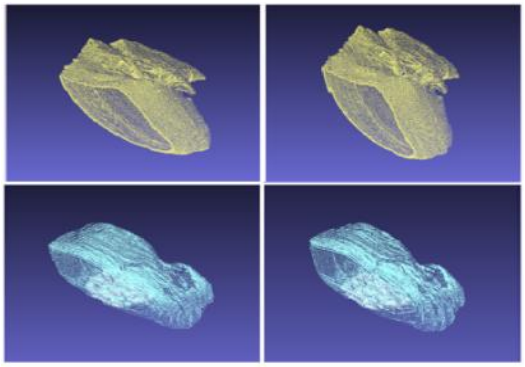


Fig.1. The 3D reconstruction results of liver and bladder with different level of scheme complexity. The results in the top row are predicted by the model with sizes of 256 Mb and 178MB, while the bottom results are predicted by the model with sizes of 65MB, 18MB.

reconstruction post-processing, since most medical image targets, such as lesion areas or biological tissues are non-rigid objects, it is necessary to consider the non-rigid SFM models to improve the accuracy of reconstruction results. So, our method is based on Non-rigid Structure from Motion (NRSfM) to improve the accuracy of reconstruction results.

Specifically, our method couples the view synthesis from the nearby frame into NRSfM model, and can obtain more accurate reconstruction results in the self-supervised learning framework. In general, our work mainly has the following contributions: (1) By transferring the pre-learning features of natural images to medical image learning, the problem of robust feature acquisition in medical image reconstruction is overcome. (2) A NRSfM-based self-supervision structure is proposed. In 3D reconstruction, a more precise optimized model can be obtained to estimate the camera structure and pose of consecutive frames. (3) A knowledge transfer based framework is designed to achieve 3D reconstruction with better time performance. A lightweight CNN is designed, as a strategy with multi-stage progressive shrinkage, which is helpful to implement the proposed method on mobile devices.

II. METHODS

In this section, we will first introduce our lightweight framework (see Fig.2).

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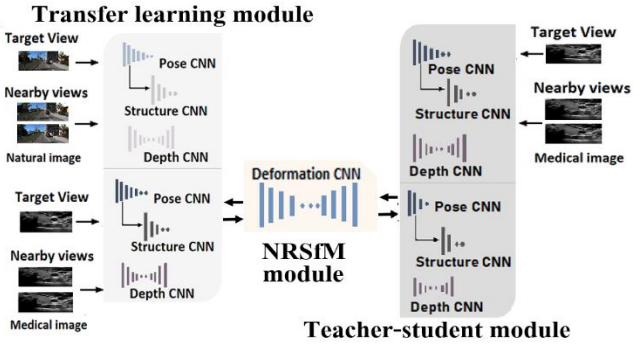


Fig.2. The overall framework. There are three modules in our framework, including the transfer learning module, the teacher-student module, and the NRSfM module.

A. Overall Framework

In our method, a knowledge transfer framework is designed to obtain better time performance in medical image construction. As shown in Fig.2, the framework has three modules, namely, the transfer learning module, the NRSfM module, and the teacher-student module in the multi-stage progressive contraction. Our lightweight model combines transfer learning with teacher-student learning. First, we use a transfer learning strategy to learn more robust features. In the transfer learning module, there are two CNN, pre-learning CNN and feature transfer CNN. Both of them have single view depth CNN, camera pose estimation CNN and camera structure estimation CNN. These CNNs are trained together. In the teacher-student module, we designed multi-level progressive learning. In order to improve the performance of CNN, we provide multiple progressive contractions and introduce a 4-level scheme example. On the basis of relevant works[24-25], we coupled deformation estimation into the SFM model and then introduced NRSfM into knowledge transfer learning. See Fig. 2.

B. NRSfM Structure for 3D Reconstruction

Our work is inspired by previous works [21-23]. we consider three consecutive source capture frames. The depth of the corresponding view is expressed as $< I_l, I_t, I_r >$. For simplicity, we will use the symbol with subscript marks of t, l and r to denote the information of the target view, its left neighborhood, and its right neighborhood respectively.

In our framework, deformation estimation CNN was designed in ResNet structure with the 6-layer encoder-decoder [26], and we coupled the prediction of deformation estimation into SFM for self-monitoring in photogrammetric knowledge learning.

C. Transfer Learning of Natural Image Knowledge

Our transfer learning strategy includes a feature pre-learning stage and a feature transfer stage.

The pre-learning sub-CNN, including the single view depth CNN and the camera pose estimation CNN, is jointly trained under the premise that the camera structure of the natural image is known. We use view synthesis loss to constrain training[21-23], which is defined as

$$L_s = \sum_p \sum_n | I_t(p) - \hat{I}_n(p) | \qquad (1)$$

where p is the pixel coordinate index, and $\hat{I}_n =< \hat{I}_l, \hat{I}_r >$, $\hat{I}_l$ and $\hat{I}_r$ denote the synthesized images projected from the left and right adjacent frames.

The camera structure annotation of the natural image data set is used as the supervision signal. Therefore, the camera structure loss can be defined as:

$$L_c = \sum_i || F_i - G_i ||_2 \qquad (2)$$

Where F_i denotes the estimated intrinsic parameters in the camera structure, and G_i is the supervision of camera structure from the natural image.

In feature transferring, the pre-learned feature from natural image is transferred to the feature fine-tuning procedure, and then, the single-view depth CNN, the camera pose estimation CNN, as well as the camera structure estimation CNN are jointly fine-tuned.

D. Teacher-student Model with Multiple Progressive Steps

In this work, we propose four-level schemes in medical image progressive learning, and design them as the multi-layer perception layers based on encoder-decoder structure. To simplify, we denote them as scheme-1, scheme-2, scheme-3, and scheme-4. The model sizes of different schemes are 281MB, 178MB, 65MB,and 18MB.Each scheme includes 4 CNNs: the single-view depth CNN, the camera pose estimation CNN, the camera structure estimation CNN, and the deformation estimation CNN. Scheme-1 has the higher complexity, and scheme-4 has the lower complexity. Every level scheme is designed as a hierarchical learning mechanism.

Both transfer learning and progressive teacher-student learning need the NRSfM-based optimization on view synthesis. Based on the recent works [21-23] with view synthesis being a supervisory signal. Taking the non-rigid deformation into consideration, we define the loss function of our model as:

$$L_{ND} = L_s + \gamma L_{f->smooth} \qquad (3)$$

Where the $L_{f->smooth}$ is the smoothness loss over the predicted 2D deformation, we minimize the L_1 norm of the second-order gradients for the predicted deformation and γ is the weight for the smoothness loss in the total loss. The optimization process of the network within one round is described in Algorithm 1.

Algorithm 1: Non-rigid Structure from Motion for 3D Reconstruction.

Input: I_l, I_t, I_r
Output: The parameter θ_1 of depth CNN, θ_2 for pose CNN, θ_3 for structure CNN.
1: **while** stopping criterion not met **do**
2: Predicting the depth of I_l, I_t, I_r by depth CNN
3: Predicting the camera parameters by structure CNN
4: Predicting the pose parameters by pose CNN
5: Predicting the 2D deformation between I_t and $\hat{I}_l$, as well as I_t and $\hat{I}_r$, respectively
6: Synthesizing target views from I_l and I_r ,and denote as $\hat{I}_l$ and $\hat{I}_r$
7: Calculating the loss function using L_{ND}
8: Updating the parameters θ_1, θ_2 and θ_3 by back propagation
9: **end while**
10: **return** θ_1, θ_2 and θ_3

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III. EXPERIMENTS

A. Datasets and Implementation

1) ***Dataset.*** In our experiment, we obtained 10 sets of ultrasound videos from **LITFL** library[27], including ultrasound scan videos of carotid artery, gallbladder, abdominal cavity, ovary, prostate, etc., and then sampled 2133 images from these videos, and finally all images are cropped to 350×400 around the medical target. The Synapse multi-organ segmentation dataset (**Synapse** [28]) , each CT volume consists of 85~ 98 slices of 512 × 512 pixels.We randomly select 10 abdominal CT scan cases, a total of 1779 axially enhanced abdominal clinical CT images, including 8 abdominal organs study. Besides, we selected 1940 pictures of 12 scenes from **Kitti dataset**[29], as the pre-training data of the model, and all frames are adjusted to a resolution of 128×416.

2) ***Implementation***. Our model is implemented in Tensorflow [30] and optimized with Adam. The learning rate for training is set to 0.0002. All images resized to 128×416 for training. The depth CNN and the pose CNN are trained jointly and the number of iterations is 40 million iterations. The structure CNN is trained about 200,000 iterations on Kitti dataset. Then fine-tune on medical image dataset. These components are jointly fine-tuned on the medical image dataset about 50,000 iterations.

B. Evaluation and Analysis

1) ***Evaluation.*** We report the average Dice Coefficient (DSC), average Jaccard Coefficient(JC), and average Hausdorff Distance (HD) on the re-projected mask. The DSC measures the intersection areas of the re-projected mask areas and ground truth areas.

2) ***Baseline*** .We select SfMLearner [21], an unsupervised scene reconstruction method for comparison. For the SfMLearner, we designed two baselines, Baseline and Baseline+BRSfM. Note that the difference between the two baselines is whether to use the NRSfM module.

3) ***Qualitative Visualization and Analysis.*** To evaluate the effectiveness of our method for 3D reconstruction, we compared our model with Baseline and Baseline+NRSfM, and show the reconstruction results of multiple views of CT image and ultrasound image. As shown in Fig.3, to make the 3D reconstruction result more intuitive, it can be seen that our proposed method can reconstruct 3D structure from 2D medical slices, which illustrates the effectiveness of our approach. The reconstruction fidelity of our proposed method is significantly better than other methods, which shows that non-rigid deformation helps CNN learn geometric structures of medical target.

4) ***Quantitative Comparisons.***We quantitatively evaluate our methods on LITFL ultrasound test set or Synapse multi-organ segmentation CT test set. Table 1 summarizes the results, it can be seen that the proposed method has better performance on several evaluation indicators, which suggests that our method achieves more accurate 3D geometric reconstruction. Compared with Baselines, the method proposed in this paper can get a higher Dice score in both LITFL ultrasound dataset and Synapse multi-organ segmentation CT dataset.

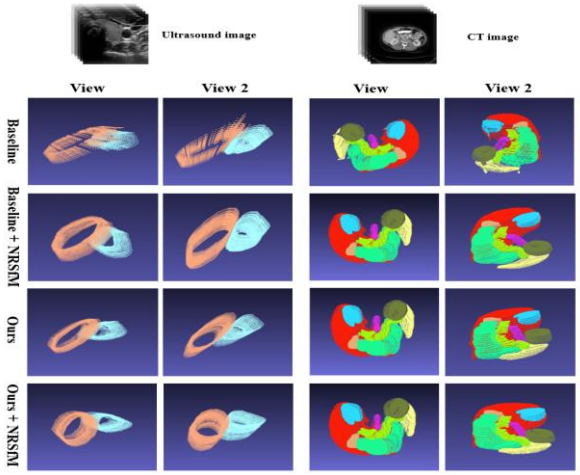


Figure 3: Results of 3D reconstruction. Figure 4: Comparison results of different methods in 3D reconstruction of ultrasound data and CT data. The NRSfM denotes Non-rigid Structure from Motion module.

TABLE 1: RESULTS OF QUANTITATIVE COMPARISON

	Ultrasound image(mean)			CT image(mean)		
	DSC(%)↑	JC(%)↑	HD↓	DSC(%)↑	JC (%)↑	HD↓
Baseline	97.2313	94.7741	0.4545	97.1141	97.1146	7.3552
Baseline + NRSfM	98.6009	97.3710	**0.1818**	97.5435	97.3418	5.8524
Ours	99.0833	98.1958	0.6363	99.1146	98.2525	6.3758
Ours *+ NRSfM*	**99.6796**	**99.3691**	0.2727	**99.8136**	**99.6289**	**5.2499**

5) ***Effectiveness of Light-weight CNN***. To test effectiveness of light-weight CNN, we compressed the model to improve the time performance of the model, so we tried to test the proposed 4-level scheme with different scale complexities as show table2. In order to prove the effectiveness of each scheme, we conducted a qualitative and quantitative analysis.

TABLE 2: PERFORMANCE OF 4-LEVEL SCHEME WITH DIFFERENT SCALE COMPLEXITIES.

	Ultrasound image(mean)			CT image(mean)		
	DSC(%)↑	JC(%)↑	HD↓	DSC(%)↑	JC (%)↑	HD↓
Ours Scheme-1	**99.6796**	**99.3691**	0.2727	99.8136	99.6289	5.2499
Ours Scheme-2	99.4527	98.9235	0.4545	95.9605	92.3224	7.9806
Ours Scheme-3	98.7538	97.6280	**0.1818**	99.2200	98.4615	2.6787
Ours Scheme-4	96.4462	93.2495	2.4752	**99.9506**	**99.9015**	**2.5000**

As shown in Fig.4, the 4-level scheme with different scale complexities we proposed can reconstruct the 3D structure from medical 2D slices, which also shows the effectiveness of the Teacher-student module we designed for the reduction model. In addition, as shown in Table 2 and Fig.4, we maintained the competitiveness of the model performance when the model size was only 18.275MB.

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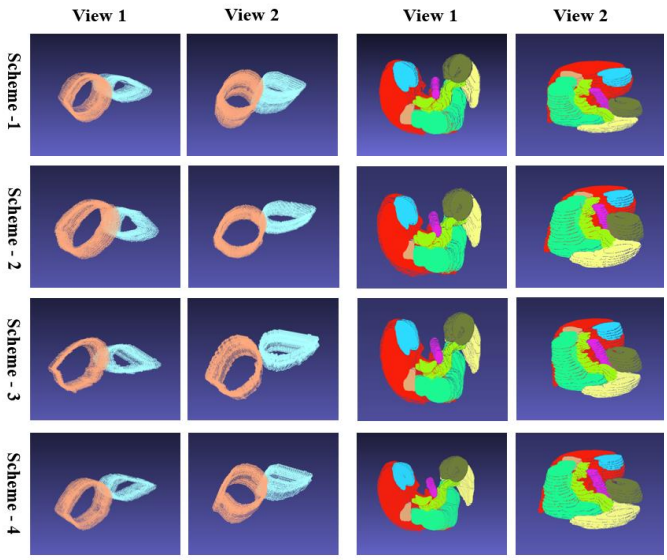


Fig.4. 3D comparison results of 4-level schemes with different scale complexity in ultrasound data and CT data.

IV. CONCLUSION

We propose a lightweight CNN based on Non-rigid Motion Structure (NRSfM) for the 3D reconstruction of medical images. We apply transfer learning to pre-learn knowledge about depth, camera structure, and pose from natural images, which helps to compensate for the lack of inherent parameters of medical sequences. we also proposed the constraint conditions of non-rigid transformation in 3D reconstruction, and designed a teacher-student training strategy. Experimental results show that our lightweight model can reconstruct 3D structures from two-dimensional medical slices.

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